

Smart Curriculum Activity & Attendance

System Architecture & Solution Design Document (SIH 2026)

1. Executive Summary

The Smart Curriculum Activity & Attendance Web Application is a lightweight, responsive Django-based prototype designed to eliminate classroom administration overhead while optimizing students' empty periods. Through QR code verification, automated timetable parsing, and dynamic resource suggestions, it converts passive free time into active learning periods.

2. Architecture & Design Patterns

The application utilizes Django's Model-View-Template (MVT) architecture for the backend. The frontend is decoupled with dynamic styling systems built using Vanilla CSS and Client-side jsQR library interfaces.

3. Core Database Tables

Table Name	Description
accounts_customuser	Stores student/teacher metadata (department, roll number, section, role).
attendance_subject	Stores academic subject titles, unique course codes, and credits.
attendance_class	Maps teachers to classrooms and enrolled students.
attendance_attendance session	UUID-identified QR attendance windows with expiration markers.
attendance_attendance record	Tracks marked attendance metrics, IP address, and marking timestamps.
timetable_timeslot	Stores weekly schedule entries, categorizing classes and free periods.
timetable_tasksuggestion	Holds academic work recommendations categorized by priority and subject.
timetable_dailyplanner	Student planner tracker logging completed tasks and custom daily notes.

4. Key Modules & Features

- **Authentication Module:** Secure role-based login interfaces redirecting users automatically to dedicated Teacher/Student analytics portals.
- **QR Code Attendance:** Automatic active session generation with countdown expirations. Students can upload or drop saved QR image files or mark directly using live webcams.
- **Timetable & Planner:** Automatically detects free periods throughout the day and auto-suggests context-aware tasks linked to active courses.
- **Analytics:** Visualization dashboard featuring line charts (daily trends), donut gauges, and a warning log flagging students falling under the required 75% attendance threshold.